Resurrecting and Extending AttnGAN

A comprehensive modernization, reproduction, and novel ablation study of fine-grained text-to-image synthesis.

State 1 (64x64) State 2 (128x128) State 3 (256x256)

DATASET: CUB-200-2011

FRAMEWORK: PyTorch 2.3 (Migrated from legacy Pytorch <1.0)

ENVIRONMENT: Kaggle T4 Dual-GPU

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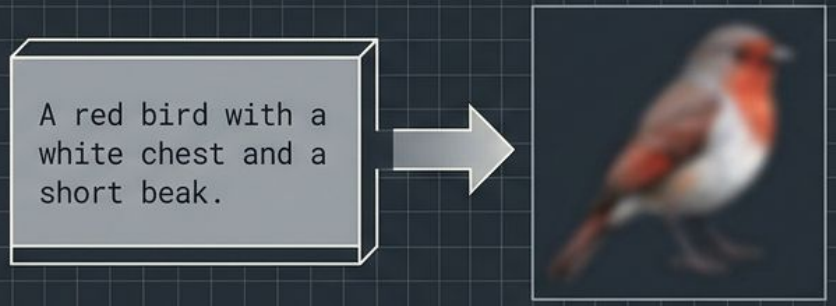
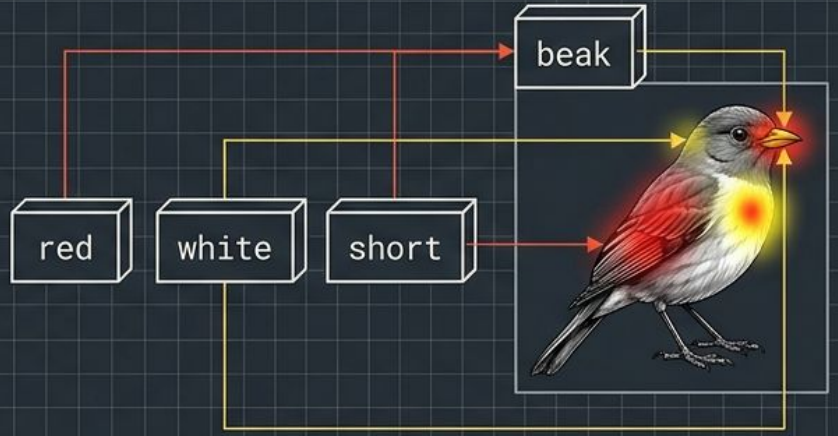
Global sentence vectors lose critical fine-grained details

The Problem: Monolithic Generation



Monolithic generation relies on a single sentence embedding. This lacks word-level information, causing semantic collapse when generating complex scenes or specific visual attributes.

The Solution: Multi-Stage Attention



AttnGAN introduces multi-stage, attention-driven refinement. The model isolates individual words and forces the generator to attend to specific sub-regions, creating a dynamic, fine-grained text-to-image mapping.

The Attentional Generative Network: Progressive Refinement

Attention Model (F_{attn1})

Attention Model (F_{attn2})

Stage 1 (G0 - 64x64)

Utilizes only the global sentence vector to sketch the primitive shape, layout, and background colors.

Stage 2 (G1 - 128x128)

Injects word-context features. Begins correcting defects and drawing specific attributes (e.g., wing color).

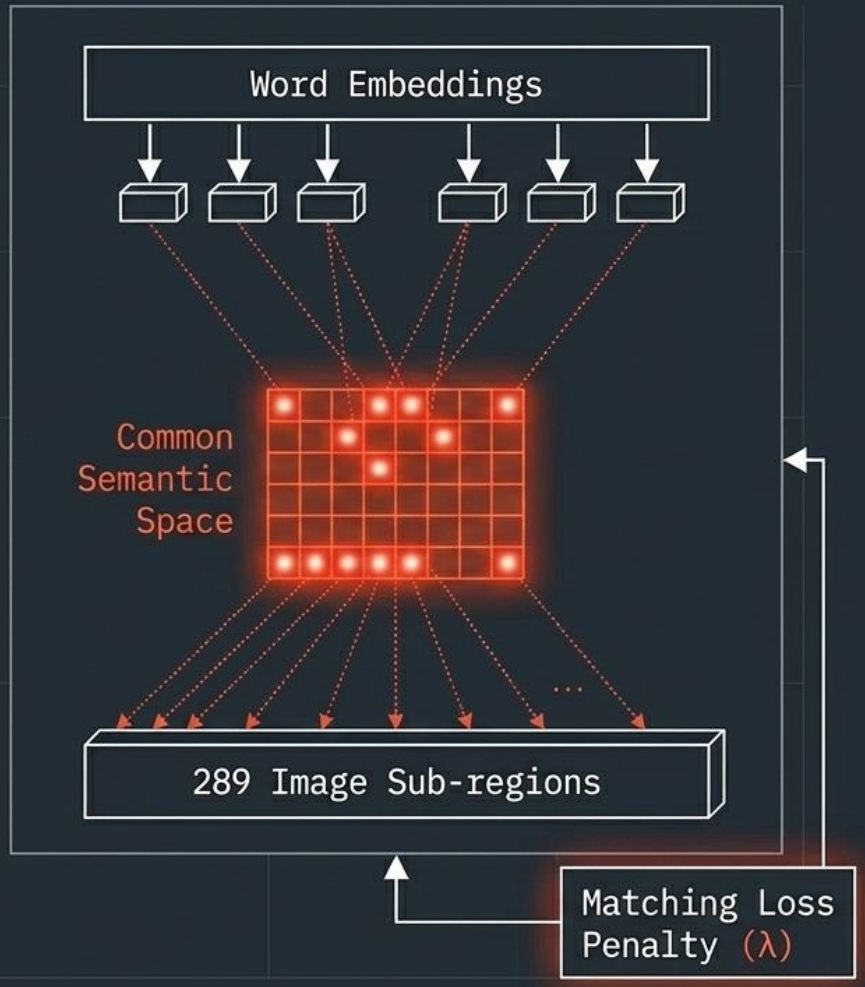
Stage 3 (G2 - 256x256)

Final refinement. Recovers omitted object details and synthesizes high-fidelity, photorealistic textures.

DAMSM: Forcing word-level semantic accountability

The Deep Attentional Multimodal Similarity Model (DAMSM) computes a fine-grained image-text matching loss.

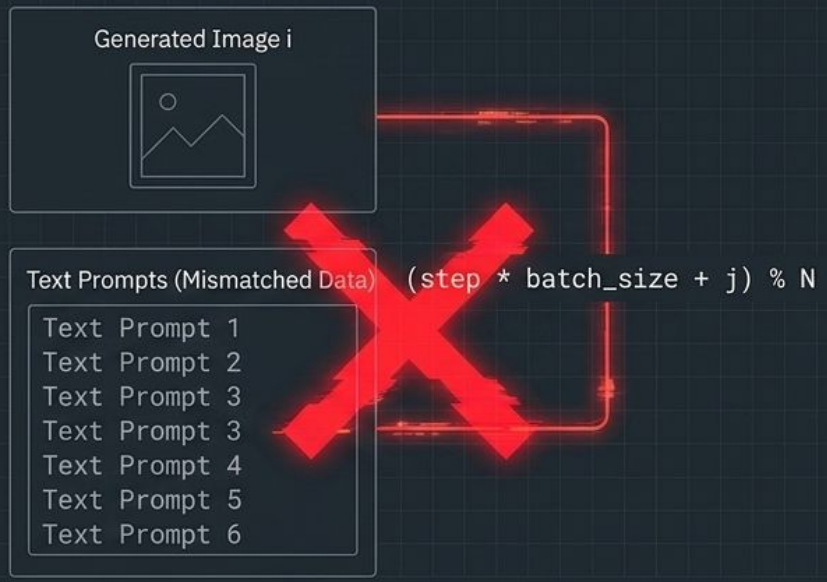
- Dual Encoders:
  - Text: Bi-directional LSTM extracts semantic vectors for every word.
  - Image: Inception-v3 extracts local visual features across 289 sub-regions.
- The network measures dot-product similarity between every word and every sub-region, heavily penalizing the generator if specific text descriptors (like 'blue crown') do not appear in the generated pixels.



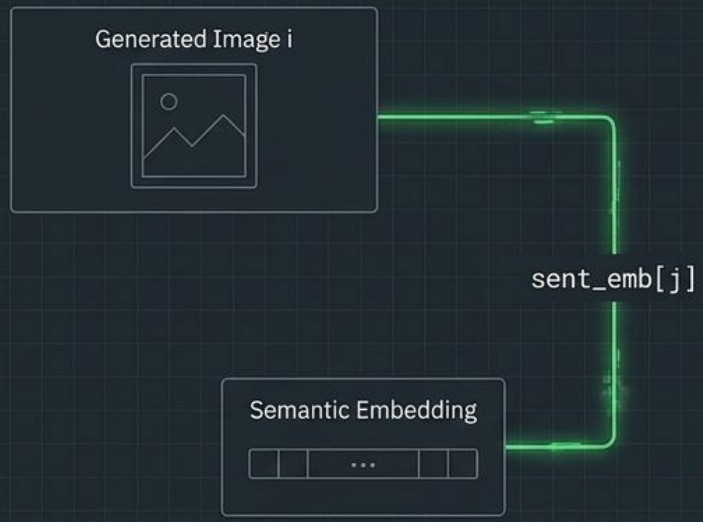
Codebase Modernization & Framework Migration

	Original 2018 Environment	Team's 2026 Environment
Language	Python 2.7	Python 3.11
Core Framework	PyTorch <1.0 Deprecated Variable and volatile flags	PyTorch 2.3 Native torch.no_grad(), unified Tensor classes
Evaluation Pipeline	Bifurcated TensorFlow for Inception Score, PyTorch for GAN	Unified PyTorch Ecosystem torch-fidelity, torch-inception-score
Execution	Multi-day scratch training	Modular feature-extraction via Kaggle T4 Dual-GPU Inference-only DAMSM loaders

Restoring evaluation integrity: The R-Precision fix



The legacy evaluation script contained a critical sampling error. Match indexing was determined by raw position, completely decoupling the generated image from its actual conditioning text and flatlining **R-Precision at ~1%**.



The evaluation loop was re-engineered to lock the true match to the specific sentence embedding used to condition that exact image in the batch, evaluated against **99 random mismatched captions**.

Baseline Validation on the CUB-200 Dataset

Before initiating novel ablations, the modernized pipeline validated the official 600-epoch pretrained AttnGAN2 model to establish an empirical ceiling. The PyTorch 2.3 migration is mathematically stable, surpassing the target Inception Score and establishing a robust baseline for controlled ablation studies.

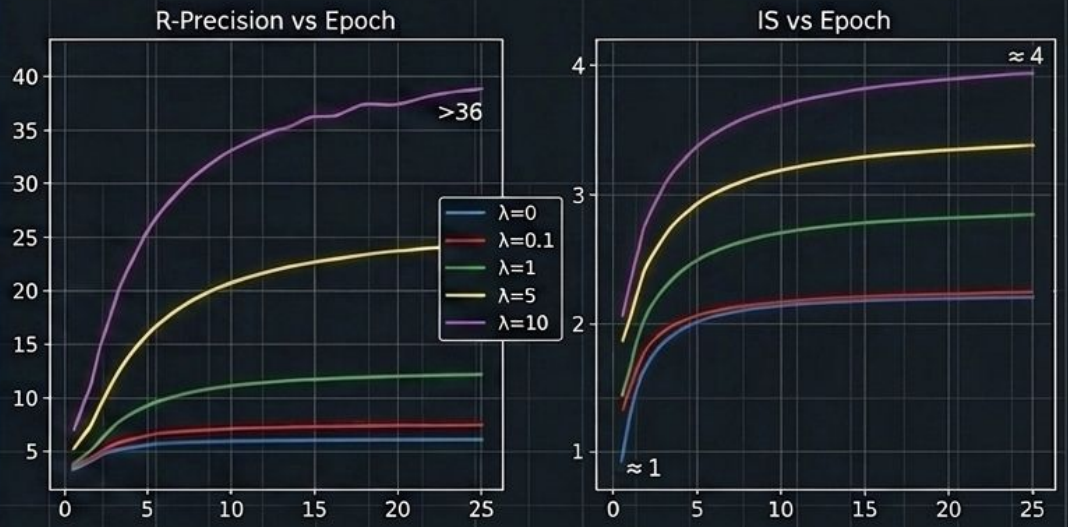
Original Paper (Target)	Our Modernized Baseline
IS ≈ 4.36	IS = 4.86
R-Precision ≈ 67.82%	R-Precision = 63.59%

Core Replication: The absolute necessity of DAMSM supervision

Experiment: Sweeping the DAMSM weight ($\lambda \in \{0, 0.1, 1, 5, 10\}$) using the AttnGAN1 configuration over 25 epochs.

The Finding: Without a strong text-alignment penalty ($\lambda \leq 0.1$), semantic fidelity completely collapses (R-Precision ≈ 6%).

The Trend: As λ increases to 10, the model is forced to respect text anchors, driving R-Precision above 36% in just 25 epochs and elevating the overall Inception Score from roughly 1 to 4.



Semantic Collapse vs. Strict Alignment (25 Epochs):

- **Semantic Collapse ($\lambda \leq 0.1$):** Model ignores text, R-Precision flatlines ≈ 6%, IS limited to ≈ 1.
- **Strict Semantic Alignment ($\lambda = 10$):** High penalty forces strong text alignment, R-Precision jumps to >36%.
- **Performance Boost:** Alignment improves quality, boosting IS from ≈ 1 to ≈ 4.

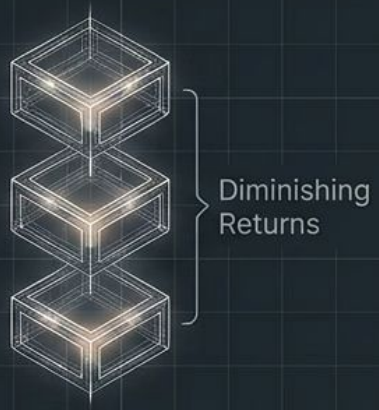
Novel Extension 1: Forcing harder choices via Attention Temperature (τ)

Mechanism: Modulating the temperature parameter (τ) scales the logits before the softmax operation in the cross-attention matrix. Lowering τ forces the network to make sharper, highly localized attention choices (e.g., locking strictly onto 'pointed beak'). For localized datasets like CUB, this diffuse-to-sharp adjustment yields strictly superior semantic grounding.

τ = 1.00 (Baseline)	→	IS: 4.86	|	R-Prec: 63.59%
τ = 0.75	→	IS: 4.93	|	R-Prec: 64.52%
τ = 0.50 (Optimal)	→	IS: 4.92	|	R-Prec: 65.16%

Novel Extension 2: The residual refinement trade-off

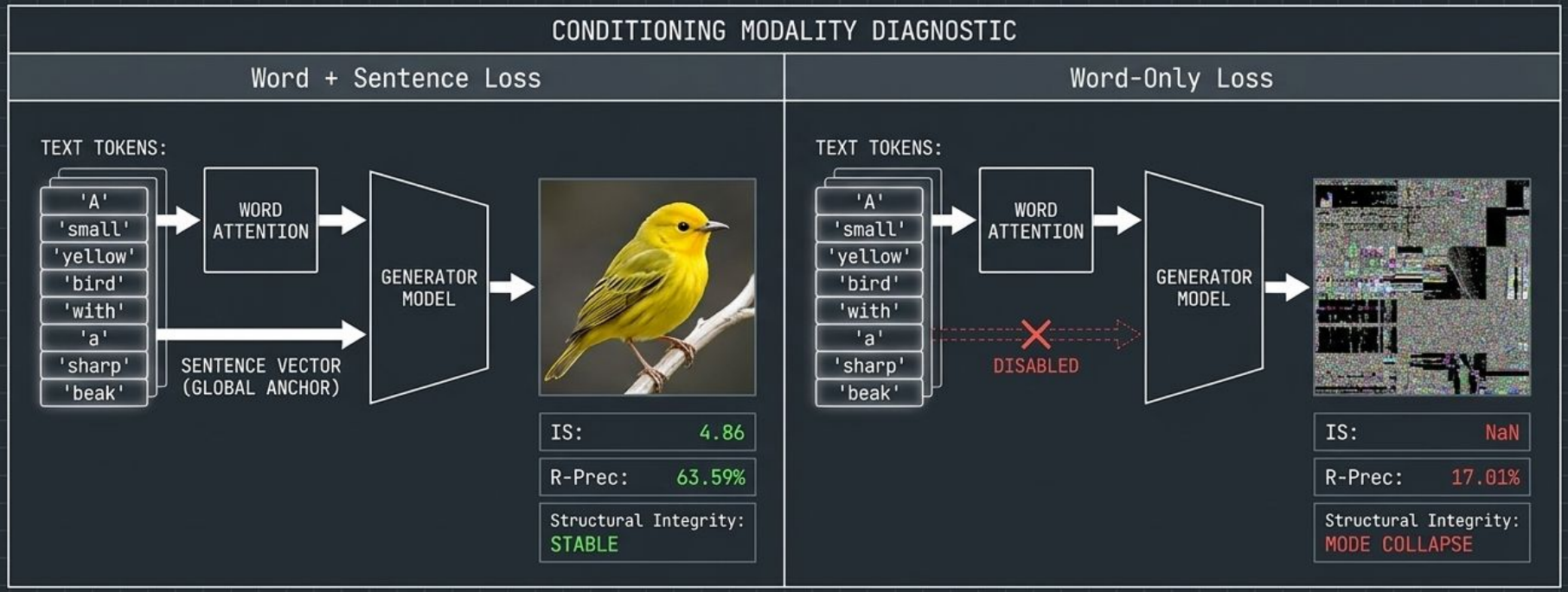
Experiment: Manipulating the core generator backbone depth by altering the number of residual blocks (r_num) in intermediate and final stages. Depth introduces a strict trade-off. Increasing from 1 to 2 blocks triggers a massive R-Precision gain (+9%), proving stronger text conditioning. However, pushing to 3 blocks over-parameterizes the network, reducing both generation diversity (IS) and semantic fidelity.

r_num = 1	OPTIMAL ZONE (Peak Semantic Accuracy) r_num = 2	r_num = 3
		
IS: 4.91 R-Prec: 68.10%	IS: 4.72 R-Prec: 77.42%	IS: 4.58 R-Prec: 76.09%

Novel Extension 3: The absolute dependency on global anchors

Experiment: Isolating the loss components by entirely disabling the global sentence-level loss, leaving only the fine-grained word-level attention.

Fine-grained details are useless without macro-spatial understanding. The global sentence embedding acts as an absolute prerequisite anchor to form a coherent base layout before word attention can refine the pixels.



Visual Evidence: Dynamic attention shifts across stages

Extracting the intermediate attention matrices proves the model dynamically shifts its cognitive focus as resolution increases.

Stage 1 (F_attn1 at 128px): Predominantly broad color descriptors to the entire body mass (e.g., locking 'blue' to the general torso).

Attn Stage 1 Top-5 words: beak: 527.9 belly: 512.6 blue: 407.3 short: 388.6 crown: 368.5

Stage 2 (F_attn2 at 256px): Shifts to pinpoint localization. It ignores broad strokes and maps specific structural nouns (e.g., 'crown', 'beak') directly to extreme edges and specific anatomical coordinates.

Attn Stage 2 Top-5 words: belly: 3822.9 beak: 2029.4 blue: 1906.7 bird: 1430.2 this: 1399.5

Synthesis: The Anatomy of Semantic Alignment

The success of AttnGAN is not merely a product of stacking layers, but of balancing four distinct counter-forces within the architecture:

1. The Macro-Anchor

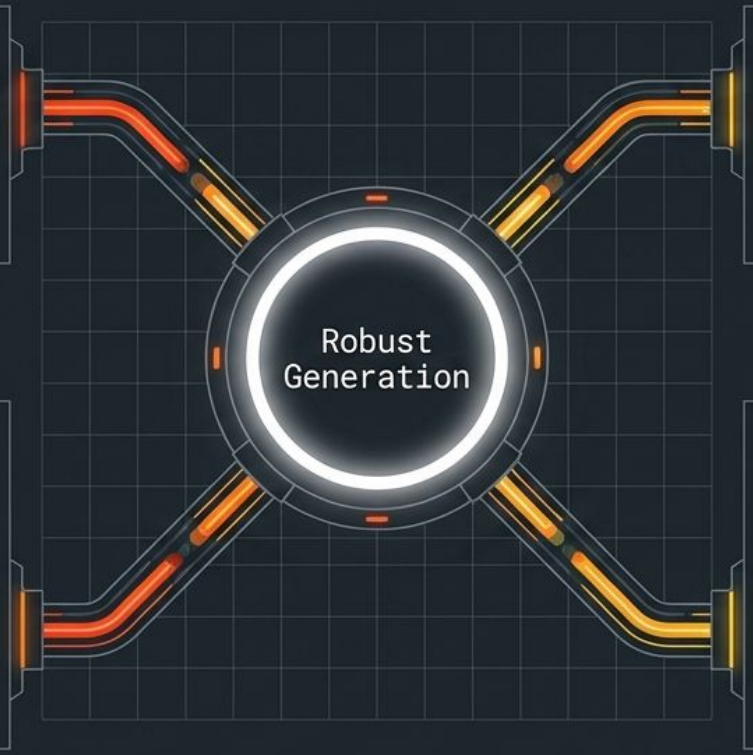
Sentence embeddings are mandatory. Word-level attention cannot create structure; it can only refine it.

Global Sentence Embedding: Required for Base Layout

2. The Attention Penalty

High DAMSM weight (λ=5+) is required to prevent the generator from collapsing into generic, unconditioned textures.

$\lambda_{_DAMSM}$ = 5.0+



3. Sharp Temperature

Tightening the softmax distribution (τ=0.5) forces superior, explicit semantic choices over diffuse guessing.

Softmax τ = 0.5 (Explicit Selection)

4. Moderate Depth

Over-parameterizing residual blocks causes diminishing returns. r_num=2 provides the optimal trade-off between image diversity and strict text alignment.

Optimal r_num = 2 (Diversity vs. Alignment)